



Exercise Set 1.2 Solution

Question 17

For which values of a will the following system have no solutions? Exactly one solution? Infinitely many solutions?

$$\begin{array}{rcl} x + 2y - & 3z = & 4 \\ 3x - y + & 5z = & 2 \\ 4x + y + (a^2 - 14)z = & a + 2 \end{array}$$

Solution:

$$\begin{array}{l} \\ R_2 - 3R_1, R_3 - 4R_1 \\ \\ R_2 - R_3 \\ \\ \frac{-1}{7}R_2 \end{array} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 3 & -1 & 5 & 2 \\ 4 & 1 & (a^2 - 14) & a + 2 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & -7 & (a^2 - 2) & a - 14 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & 0 & (a^2 - 16) & a - 4 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & 1 & -2 & 10/7 \\ 0 & 0 & (a^2 - 16) & a - 4 \end{array} \right]$$

The Gauss-Jordan process will reduce this system to the equations

$$\begin{array}{l} x + 2y - 3z = 4 \\ y - 2z = 10/7 \\ (a^2 - 16)z = a - 4 \end{array}$$

If $a = 4$, then the last equation becomes $0 = 0$, and hence there will be infinitely many solutions-for instance,

$$z = t, y = 2t + \frac{10}{7}, x = -2 \left(2t + \frac{10}{7} \right) + 3t + 4$$

. If $a = -4$, then the last equation becomes $0 = -8$, and so the system will have no solutions.

Any other value of a will yield a unique solution for z and hence also for y and x .

Exercise Set 1.6 Solution

Question 10

Solve the three systems in Exercise 9 using the method of Example 2.

$$\left[\begin{array}{ccc|c|c|c} 1 & 2 & 3 & -1 & 5 & -1 \\ 1 & -1 & 1 & 3 & 0 & -1 \\ 1 & 1 & 0 & 4 & 0 & 3 \end{array} \right] \xrightarrow{R_2-R_1, R_3-R_1} \left[\begin{array}{ccc|c|c|c} 1 & 2 & 3 & -1 & 5 & -1 \\ 0 & -3 & -2 & 4 & -5 & 0 \\ 0 & -1 & -3 & 5 & -5 & 4 \end{array} \right]$$

$$\left[\begin{array}{ccc|c|c|c} 1 & 2 & 3 & -1 & 5 & -1 \\ 0 & -3 & -2 & 4 & -5 & 0 \\ 0 & -1 & -3 & 5 & -5 & 4 \end{array} \right] \xrightarrow{R_2 R_3} \left[\begin{array}{ccc|c|c|c} 1 & 2 & 3 & -1 & 5 & -1 \\ 0 & -1 & -3 & 5 & -5 & 4 \\ 0 & -3 & -2 & 4 & -5 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c|c|c} 1 & 2 & 3 & -1 & 5 & -1 \\ 0 & -1 & -3 & 5 & -5 & 4 \\ 0 & -3 & -2 & 4 & -5 & 0 \end{array} \right] \xrightarrow{R_1+2R_2, R_3+3R_2} \left[\begin{array}{ccc|c|c|c} 1 & 0 & -3 & 9 & -5 & 7 \\ 0 & -1 & -3 & 5 & -5 & 4 \\ 0 & 0 & -11 & 19 & -20 & 12 \end{array} \right]$$

$$\left[\begin{array}{ccc|c|c|c} 1 & 0 & 0 & 4.28 & -0.71 & 2.71 \\ 0 & 1 & 0 & -0.28 & 0.71 & 0.28 \\ 0 & 0 & 1 & -1.57 & 1.42 & -1.42 \end{array} \right]$$

Question 16

Find conditions that the b's must satisfy for the system to be consistent.

$$6x_1 - 4x_2 = b_1$$

$$3x_1 - 2x_2 = b_2$$

Solution:

$$\left[\begin{array}{cc|c} 6 & -4 & b_1 \\ 3 & -2 & b_2 \end{array} \right] \xrightarrow{R_2 \leftrightarrow 2R_1} \left[\begin{array}{cc|c} 3 & -2 & b_2 \\ 6 & -4 & b_1 \end{array} \right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc|c} 3 & -2 & b_2 \\ 0 & 0 & b_1 - 2b_2 \end{array} \right]$$

One can see, for consistent second row has to get completely zeros, which deduce $b_1 = 2b_2$.

Question 23

Since $Ax = \mathbf{0}$ has only $x = \mathbf{0}$ as a solution, Theorem 1.6.4 guarantees that A is invertible. By Theorem 1.4.8 (b), A^k is also invertible. In fact,

$$(A^k)^{-1} = (A^{-1})^k$$

Since the proof of Theorem 1.4.8 (b) was omitted, we note that

$$\underbrace{A^{-1}A^{-1}\cdots A^{-1}}_{\substack{k \\ \text{factors}}} \underbrace{AA\cdots A}_{\substack{k \\ \text{factors}}} = I$$

Because A^k is invertible, Theorem 1.6.4 allows us to conclude that $A^k x = \mathbf{0}$ has only the trivial solution.

Question 24

Let $A\mathbf{x} = \mathbf{0}$ be a homogeneous system of n linear equations in n unknowns, and let Q be an invertible $n \times n$ matrix. Show that $A\mathbf{x} = \mathbf{0}$ has just the trivial solution if and only if $(QA)\mathbf{x} = \mathbf{0}$ has just the trivial solution.

Proof: First let $A\mathbf{x} = \mathbf{0}$ holds. Now we apply Q matrix from L.H.S we will have

$$\begin{aligned} Q(A\mathbf{x}) &= Q\mathbf{0} \\ (QA)\mathbf{x} &= \mathbf{0} \quad \because \text{Assoociative property} \end{aligned}$$

Now we let $(QA)\mathbf{x} = \mathbf{0}$ and we apply Q^{-1} from L.H.S because Q is invert-able we will have

$$\begin{aligned} Q^{-1}(QA)\mathbf{x} &= Q^{-1}\mathbf{0} \\ (Q^{-1}Q)A\mathbf{x} &= Q^{-1}\mathbf{0} \quad \because \text{Assoociative property} \\ IA\mathbf{x} &= \mathbf{0} \\ A\mathbf{x} &= \mathbf{0} \end{aligned}$$

Question 25

Suppose that x_1 is a fixed matrix which satisfies the equation $Ax_1 = \mathbf{b}$. Further, let x be any matrix whatsoever which satisfies the equation $Ax = \mathbf{b}$. We must then show that there is a matrix x_0 which satisfies both of the equations $x = x_1 + x_0$ and $Ax_0 = \mathbf{0}$. Clearly, the first equation implies that

$$x_0 = x - x_1$$

This candidate for x_0 will satisfy the second equation because

$$Ax_0 = A(x - x_1) = Ax - Ax_1 = \mathbf{b} - \mathbf{b} = \mathbf{0}$$

We must also show that if both $Ax_1 = \mathbf{b}$ and $Ax_0 = \mathbf{0}$, then $A(x_1 + x_0) = \mathbf{b}$. But

$$A(x_1 + x_0) = Ax_1 + Ax_0 = \mathbf{b} + \mathbf{0} = \mathbf{b}$$

Question 26

Solution:

For $B = A^{-1}$, we use part (a)

If B is a square matrix satisfying $BA = I$, then $B = A^{-1}$

Given: $AB = I$ and $BA = I$.

Hence $AB = BA = I$.

Hence $B = A^{-1}$ for $AB = I$.

Another fact can be seen $A^{-1}A = AA^{-1} = I$.